RS2B Reader Study v2 — Cohort & Results Report

Generated: 2026-08-29   Filter: ≥95% answered items per test   Cohort N = 71

Narrative

Study design. RS2B is a prospective sequential multimodal reader study evaluating whether DermFM-Zero (zero-shot) assistance improves diagnostic accuracy and management appropriateness in skin cancer specialist care. Each reader evaluated paired dermoscopic + clinical image cases spanning 11 lesion classes suspected of malignancy. For every case a reader produced (i) an unaided differential diagnosis and clinical management decision, then (ii) repeated the assessment with DermFM-Zero's top-3 probabilities visible. One test = 15 cases × 2 modes = 30 readings.

v2 changes vs the original Nature submission (34 readers):
- Expanded cohort — the clinician's updated export contains 87 registered participants (all 34 v1 readers retained, plus 53 new).
- New management appropriateness table — data-driven 11-class × 4-action mapping (data/management_table.csv). Replaces the hard-coded dict in v1 and now matches the clinician's lme4 GLMM script.
- New inclusion rule — keep a test only if ≥95% of its 30 readings are valid (i.e., ≥29 valid answers, no timeouts, no blanks). Selects 71 readers / 165 tests / 4,929 readings for the analytic cohort.

Statistical analysis. Reader-level proportions are compared with paired t-tests (this folder). A separate folder (R2B_glmm_analysis/) fits a binomial generalised linear mixed model adjusting for class difficulty and within-reader / within-test correlation, used as the primary inference in the manuscript.

Headline results (paired t-test, 71 readers)

Metric	Unaided	AI-assisted	P (paired t)
Overall diagnostic accuracy	51.4%	59.4%	<0.001
Overall management appropriateness	67.8%	70.4%	0.00856
Non-expert accuracy	49.1%	58.6%	<0.001
Expert accuracy	53.2%	60.0%	<0.001
Non-expert mgmt appropriateness	67.3%	69.7%	0.139
Expert mgmt appropriateness	68.1%	70.9%	0.0258

Tip: For the primary inference in the manuscript, use the GLMM outputs from the sibling folder R2B_glmm_analysis/ — they adjust for image difficulty and within-reader correlation.

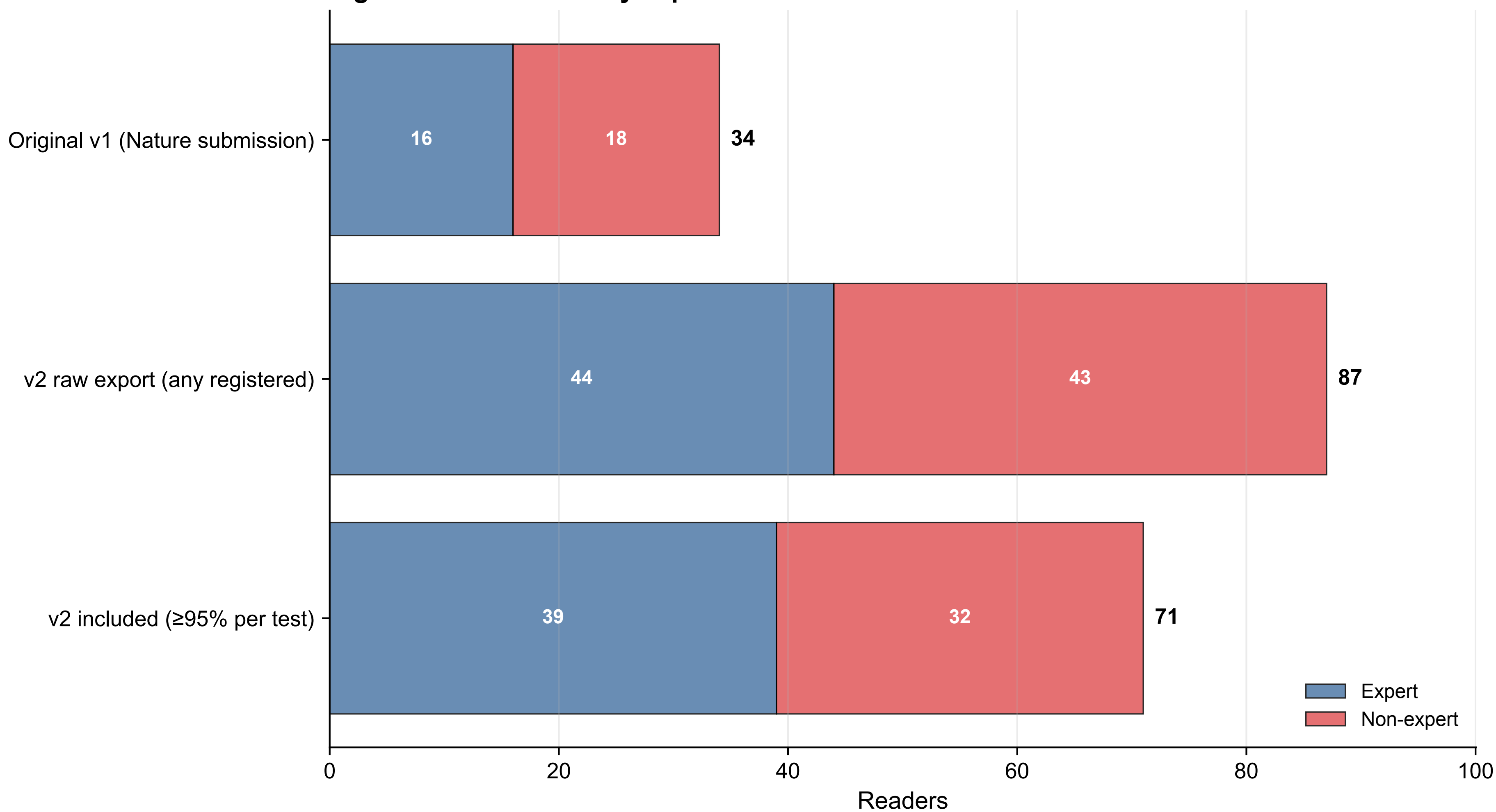
Page 2 · Reader cohort summary

All 34 original readers were retained in v2; the ≥95% filter preserves 33 of them and adds 38 new participants for a final cohort of 71 (39 expert, 32 non-expert).

Table 1. Cohort progression

Cohort	N readers	Experts	Non-experts	Shared with v1	New vs v1
Original v1 (Nature submission)	34	16	18	34	0
v2 raw export (any registered)	87	44	43	34	53
v2 included (≥95% per test)	71	39	32	33	38

Figure 1. Cohort size by expertise



Page 3 · Demographics of the included 71-reader cohort

Profession is the most informative proxy for clinical experience; the IRT ability score (expected_accuracy_2pl, 0–1) is the platform-derived continuous skill estimate.

Table 2. Profession distribution

Profession	N	%
dermatologyResident	23	32.4
boardCertifiedDermatologist	21	29.6
generalPractitioner	9	12.7
medicalStudent	6	8.5
resident	4	5.6
medicalSpecialist	3	4.2
physicianAssistant	2	2.8
nonMedical	2	2.8
nursePractitioner	1	1.4
TOTAL	**71**	**100.0**

Table 3. Gender

Gender	N	%
Unknown	71	100.0
TOTAL	**71**	**100.0**

Table 4. Age band

Age band	N	%
Unknown	71	100.0
TOTAL	**71**	**100.0**

Table 5. Expertise group × IRT ability score

Expertise	N	Mean	Median	Min	Max
expert	39	0.766	0.772	0.699	0.874
non-expert	32	0.570	0.594	0.273	0.689

Page 4 · Inclusion / exclusion flow

Table 6. Reader inclusion flow

Step	N	Share
Total readers in raw export	87	—
Total tests in raw export (one test = 15 cases × 2 modes)	194	—
Tests kept after ≥95% filter	165	(85%)
Tests excluded (too many timeouts / blanks)	29	(15%)
Readers retained (at least one valid test)	71	(82%)
Readers excluded (no valid test)	16	(18%)

Filter rule (matches new R script)

A test (one reader's 15-case round in both Unaided and AI modes) is included if at least 95% of its 30 expected answer slots are valid — i.e., ≥29 of 30 readings are non-blank, non-timeout. Tests below this threshold are dropped wholesale. A reader is retained if at least one of their tests survives. Readers who completed multiple test sessions contribute every kept session.

Strict alternative (not used in this folder). Earlier versions of the pipeline required all 30 slots filled (100%). That rule kept 62 readers / 144 tests and produced quantitatively similar headline results. The strict-15 version is still available in the GLMM sensitivity folder.

Page 5 · Results summary tables

Table 7. Overall results (Unaided vs AI)

Outcome	Condition	Mean	95% CI	P
Accuracy	Without AI assistance	51.4%	[0.484, 0.544]	—
Accuracy	With AI assistance	59.4%	[0.568, 0.621]	<0.001
Mgmt appropriate	Without AI assistance	67.8%	[0.654, 0.701]	—
Mgmt appropriate	With AI assistance	70.4%	[0.681, 0.726]	0.00856

Table 8. Stratified by experience

Outcome	Group	Unaided	AI	P	P (Bonf)	N
Accuracy	Non-expert	49.1%	58.6%	<0.001	<0.001	32
Accuracy	Expert	53.2%	60.0%	<0.001	0.0013	39
Mgmt appropriate	Non-expert	67.3%	69.7%	0.139	0.279	32
Mgmt appropriate	Expert	68.1%	70.9%	0.0258	0.0517	39

Table 9. Class-specific accuracy

Class	Unaided	AI	P	P (Bonf)	N (Una.)	N (AI)
NV	56.3%	62.7%	0.0616	0.677	71	71
MEL	66.2%	67.5%	0.724	1	71	71
BKL	34.4%	46.5%	<0.001	0.001	71	71
BCC	61.1%	71.7%	0.00191	0.021	71	71
AKIEC	48.2%	58.5%	0.0429	0.472	71	71
DF	49.7%	68.0%	<0.001	<0.001	70	70
INF	19.9%	29.6%	0.114	1	40	40
VASC	75.0%	71.6%	0.508	1	71	71
SCCKA	39.3%	47.4%	0.0973	1	70	70
OTHER_BEN	17.3%	25.1%	0.134	1	47	47
OTHER_MAL	0.0%	0.0%	—	—	17	17